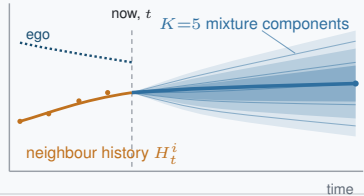


→ forward pass (deployment and Stage 2 training)

- - -> Stage 2 gradient (training only)

Trajectory predictor f_θ

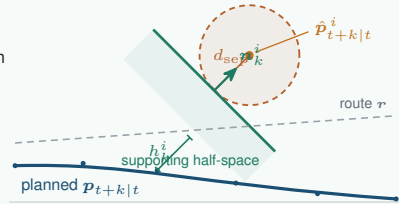


mixture mean

$$\hat{\mathbf{p}}_{t+k|t}^i$$

Planning layer QP ϕ

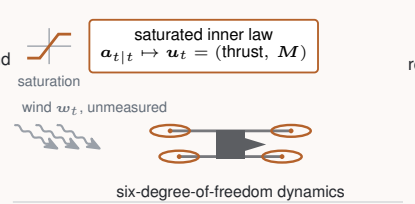
horizon H_p



first command

$$\mathbf{a}_{t|t}$$

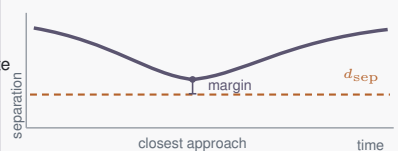
Vehicle and inner law



realized state

$$\mathbf{x}_t$$

Operational outcome



- Φ_{los} loss of separation, any-time per episode
- Φ_{dev} terminal lateral deviation
- Φ_{effort} control activity

Split-conformal calibration

empirical radius about the mixture mean
analysis only; not in the control path

state feedback, every control step

Stage 2: $\nabla_\theta \mathcal{L}_{\text{TASL}}$ propagated through F , the inner law, and the QP; planner configuration ϕ held fixed

$$\mathcal{L}_{\text{TASL}} = \lambda_s \tilde{\Phi}_{\text{los}} + \lambda_d \Phi_{\text{dev}} + \lambda_u \Phi_{\text{effort}} + \lambda_a \mathcal{L}_{\text{ADE}}$$